1] What is deep learning?

Deep learning is a subset of Machine learning that uses multi-layered neural networks to learn complex patterns from data.

2] What is a Neural Network?

A neural network is a system inspired by the human brain made of input layer, hidden layers & output layer. Each node = neuron that processes data using weights

3] Difference b/w ML & DL

Feature	Machine learning	Deep learning
Data	Small / Medium	Large
Feature engineering	Manual	Automatic
Model complexity	Low	High
Hardware	CPU	GPU
Example	Linear regression	CNN, RNN

4] What is forward propagation?

Data flows from input → hidden layers → output

Model makes prediction

5] What is Backward propagation?

Error is propagated backwards to update weights using gradients

6] What is gradient descent ?
It is an optimization algorithm used to minimize loss
It updates weights like : $w = w - \alpha \cdot \text{gradient}$

7] What are types of gradient descent ?
- Batch Gradient Descent
- Stochastic Gradient Descent (SGD)
- Mini-batch Gradient Descent

8] What is activation function?
It adds non-linearity to the model so it can learn complex patterns.

9] What are types of activation function?
- Sigmoid function, $\sigma(x) = \frac{1}{1+e^{-x}}$
- Tanh, $\tanh(x) = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}}$
- ReLU, $f(x) = \max(0, x)$
- Leaky ReLU, $f(x) = \begin{cases} x & \text{if } x > 0 \\ 0.01x & \text{if } x \leq 0 \end{cases}$
- Softmax function, $s(x_i) = \frac{e^{x_i}}{\sum e^{x_j}}$

10] What are features of deep learning?

- Automatic feature extraction
- Handles large data
- High accuracy
- Works well for unstructured data (image, text)

11] What is Loss function?

Measures how ~~the~~ wrong the prediction is.

Examples : MSE - Mean Squared Error (regression)

12] What is optimization?

It is the process of minimizing loss using algorithms like Gradient Descent, Adam, RMSProp

13] What is perceptron?

It is the simplest neural network (single neuron)

It can solve linear problems only.

14] What is Multi-layer Perceptron?

It has multiple hidden layers.

It can solve non-linear problems.

15] Difference b/w Gradient Descent & SGD

Feature	Gradient Decent	SGD
Data	full dataset	One sample
Speed	slow	Fast
Accuracy	stable	Noisy

16] What is Adam?

- Adam = Adaptive Moment Estimation
- It combines momentum & RMSProp
- It is fast, effecient & widely used.

- Adam adjusts the learning rate automatically for each parameter & learns faster & more effeciently

1. Tracks the avg of gradients
2. Tracks the squared gradients
3. Adapts learning rate for each parameter individually.

17] What is ANN?

- ANN (Artificial Neural Network) is a ML model inspired by the human brain.
- Each neuron applies weights, bias & an activation function to learn patterns
- Learning happens by adjusting weights using back-propagation + optimization (like Adam).

When to use ANN?

- Complex patterns exist
- Large dataset is available
- Feature Engineering is hard
- Applications like image recognition, speech recognition, NLP, recommendation systems,

Advantages of ANN :-

- Handles non-linearity
- Automatic feature learning
- High accuracy
- Adaptability
- Works across domains

18] What is CNN? (Convolution Neural Network)

- It is a type of deep learning model mainly used for image processing.
- Detects patterns like edges, shapes, textures

Ex: Identifying whether an image contains a cat or d

19] What is RNN? (Recurrent Neural Network)

- It is designed for sequential data.
- It remembers previous inputs using loops & used when order matters

Appl'n :- - Text prediction
Time-series forecasting

20] What is LSTM ?

- LSTM (Long short-term memory) is a special type of RNN that solves memory problems
- It can remember long-term dependencies

Ex: Predicting next word in a sentence with long context

21] What is GRU?

- GRU (Gated Recurrent Unit) is a simpler version of LSTM
- It is faster & less complex.
- works similarly to LSTM but with fewer parameters

22] What is GAN?

- GAN (Generative Adversial Network) generates new data similar to real data
- Two networks compete:
  - Generator → creates fake data
  - Discriminator → detects real vs fake

23] What is transformer?

- It is a modern deep learning architecture used in NLP.
- It uses attention mechanism instead of sequence loops.

24] What is overfitting?

- Model learns too much (including noise)
- High accuracy on training data, poor on new data
- Model too complex

25] What is underfitting?

- Model learns too little.
- Poor performance on both training & testing
- Model too simple.